KERALA SSLC | CHEMISTRY

METALS

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

(A) Various types of steels are used for industrial purposes.

(a) Write the steel that can be used for the given purposes.

(i) For the manufacture of springs, knives and drills.

(ii) Construction of railway tracks and rafters.

(b) Write one use of electric steel.

OR

(B) A very dilute sodium chloride solution is taken in two test tubes. Put an iron nail wrapped in copper wire in one test tube and an iron nail wrapped in magnesium wire in the other test tube. Keep the test tubes for few days.

(a) In which test tube does the nail undergo rusting ? Write the reason.

(b) Although aluminium is a very reactive metal, it resists corrosion to a certain extent. Why ?

---

2. 2026 **2 MARKS**

Zinc blende (ZnS) is an ore of zinc.

(a) Which is the concentration method suitable for zinc blende ?

(b) Which is the process used to convert the concentrated ore into its oxide ?

---

3. 2026 **1 MARK**

Some statements related to the manufacture of aluminium are given.

(i) The ore is treated with hot NaOH solution.

(ii) Electricity is used as the reducing agent to extract the metal.

(iii) As a result of electrolysis, aluminium is obtained at the anode.

Which of the following is correct ?

(a) Statements (i) and (iii) are correct, but (ii) is not correct.

(b) Statements (i) and (ii) are correct, but (iii) is not correct.

(c) Statements (ii) and (iii) are correct, but (i) is not correct.

(d) Statements (i), (ii) and (iii) are correct.

---

4. 2025 **4 MARKS**

Iron is a very important metal in the industrial and construction field.

(a) Write the name and chemical formula of ore of iron used in the industrial production of iron.

(b) Write the chemical equation of slag formation in the industrial production of iron.

(c) Which mineral of iron is known as fool's gold ?

(d) Write the names of two alloy steels which contain the same constituent elements.

5. 2025 **2 MARKS**

Two reactions related to extraction of metals are given. Among these reactions identify calcination and roasting.

(a) Cu_2S ore is converted to Cu_2O by heating.

(b) Carbonates and hydroxides of metals decompose to form their oxides.

6. 2025 **1 MARK**

Which of the given compounds is a mineral of Aluminium ?

(Calamine, Cuprite, Cryolite, Magnetite)

7. 2024 **3 MARKS**

Aluminium is the most abundant metal on the earth's crust.

(a) Which is the ore of aluminium ?

(b) Describe the different steps in the concentration of this ore.

8. 2024 **2 MARKS**

Stainless steel, Nichrome and Alnico are alloy steels.

(a) Which alloy steel is used for the production of heating coils ?

(b) Identify the alloy steels having the same components among these.

9. 2024 **1 MARK**

The ore of a metal is lighter than the impurities. Which method is suitable for its concentration ?

(Levigation, Froth floatation, Magnetic Separation, Leaching)

10. 2022 **4 MARKS**

Haematite is converted into iron by reactions taking place in blast furnace.

(a) Write the molecular formula of Haematite.

(b) Which substance acts as the reducing agent in this process ?

(c) Molten iron is produced along with slag from the furnace. What is meant by slag ?

(d) Write the chemical equation that shows the formation of slag.

11. 2022 **3 MARKS**

(a) Molten iron obtained from the blast furnace contains 4% carbon and other impurities. What is this known as ?

(b) Which alloy steel is used for making permanent magnets ?

(c) Some alloy steels contain the same component. Then how do they possess different properties ?

12. 2022 **1 MARK**

In which electrode aluminium metal is produced during the electrolysis of Alumina ?

13. 2022 **1 MARK**

Find the relation and fill up suitably.

(a) Tin stone : Magnetic separation

(b) Bauxite : _____

14. 2021 **4 MARKS**

Identify the suitable concentration method from the box and then complete the table.

Leaching, Froth floatation, Magnetic Separation, Levigation

Properties of ores	Properties of the impurities present in the ore	The method of Concentration
High density	Low density	(a)
Magnetic in Nature	Non-magnetic in nature	(b)
Lighter Sulphide ores	High density	(c)
Aluminium ores that get dissolved in solution	Insoluble in the same solution	(d)

15. 2021 **3 MARKS**

Some chemical reactions taking place in blast furnace are given.

(i) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

(ii) $CaCO_3 \rightarrow CaO + CO_2$

(iii) $CaO + SiO_2 \rightarrow CaSiO_3$

(a) What is the chemical formula of haematite ?

(b) Which compound reduces haematite ?

(c) Identify the flux used here.

16. 2021 **3 MARKS**

Some metals and their refining methods are given. Find the appropriate pair.

Metal : Tin, Copper, Zinc

Refining Method : Distillation, Liquation, Electrolytic Refining

17. 2021 **2 MARKS**

Two methods are used to convert concentrated ore into its oxide.

(a) Calcination (b) Roasting

What is the difference between these ?

18. 2021 **1 MARK**

Metals like gold, platinum are found in the native state in the earth's crust. Why ?